

Geometric Calibration with Multi-Viewpoints for Multi-Projector Systems on Arbitrary Shapes Using Homography and Pixel Maps

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ABSTRACT

We propose a geometrical calibration method for the MISATO astronomical observatory in Japan. The primary objective of our projection system is enabling a large-scale geometrical projection calibration for near-planner arbitrarily-shaped ground projection with a multi-projector system via temporally-placed camera capturing. The obtained results exhibited an average distortion reduction of 84.7% less than the simple homography-based method.

Keywords: Projection mapping, homography, pixel maps.

Index Terms: Computer graphics

1 INTRODUCTION

A whole-field starry sky projection of a counter hemisphere onto the ground offers an immersive experience comparable to floating in the space. In this study, we propose a low-effort geometrical calibration method for the large-scale tiled projection mapping of the MISATO astronomical observatory in Japan (Figure 1). The ground is approximately flat with a gentle curve, jump edge, dent, and other uneven features. Therefore, pioneering approaches that assume projection on a planer surface (e.g., [1]) are unsuitable.

The proposed system comprises five projection units hung on a rail (one or two projectors per unit; nine projectors in total.), which cast starry sky images on a near location to a far location in each direction centered on the tower. Because these projection units are fixed on the observation deck, an ad hoc geometric calibration [2] is not required. However, vibrations triggered by walking galleries and strong winds cause misalignment and may require daily geometrical calibrations for the tiled projection. The main challenge for our system is that it cannot adopt a permanently installed camera because the projection units are aligned with the projection contents in each season. Another challenge is that a single camera cannot acquire the entire large-scale projection using nine projectors, as illustrated in Figure 2. Hence, our calibration procedure employs a temporally placed camera.

Recently, several automatic multi-projector geometry calibration methods for nonplanar surfaces have been proposed. However, these methods assume special shape projection surfaces, such as a sphere [3], cylinder [4], and other smooth curve shapes presented with distortion models [5]. However, none of these are applicable to practical shapes, including step and roof edge. Our contribution in this study is enabling a large-scale geometrical projection calibration for the multi-projector system of near-planner arbitrarily-shaped ground projection via temporally-placed camera capturing.

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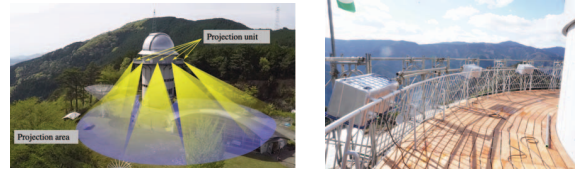


Figure 1: MISATO astronomical observatory

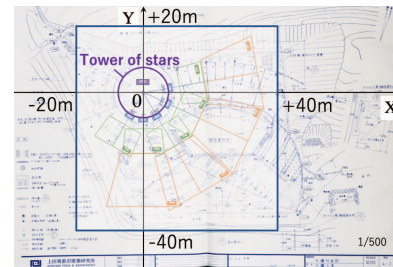


Figure 2: Definition of the world coordinate system

2 PROPOSED METHOD

We consider the camera capture as the viewer's field of view (FOV), and a camera is positioned for each viewing position (four positions for the 120° horizontal viewing angular span). Therefore, the projection is seamlessly connected to the FOV of each viewer. With this calibration, our system enables large-scale geometric projection calibrations for multiple viewpoints.

The proposed geometric projection calibration from multiple viewpoints is achieved in the following four steps (Figure 3):

1. Design the projection image on the world coordinate system
2. Generate desired camera images
3. Create the projection images with a unified coordinate
4. Correct distortions using homography and pixel maps

2.1 Desired camera image generation

The desired camera image at the reference position is obtained by clipping the designed projection image using the homography obtained by four marks of the position of the laser-ink marker falling on the stakes. Each stake location defines X and Y, while the sheet laser using the laser-ink marker defines Z = 0. For the other three desired camera images, the correspondence with the world coordinates was obtained by the homography between the camera images. Camera-to-camera homographies were calculated by sampling the pixel maps $C_k 2P_j$ and $C_l 2P_j$.

2.2 Unifying coordinate systems using homography

The coordinate systems of multiple projectors are unified into a single projector coordinate system using the homography between projectors via camera capture. The homography $H_{P_i 2P}$ between the

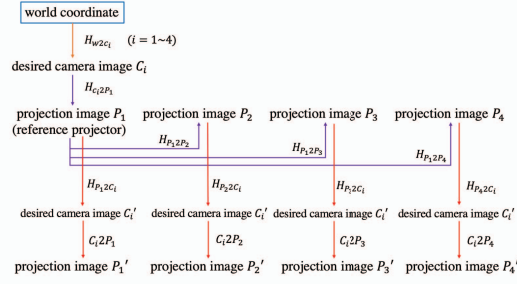


Figure 3: Flow of the proposed method

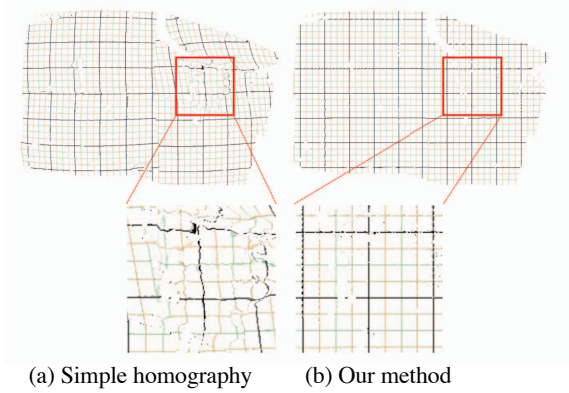


Figure 4: Simulation results on temporal camera view

reference projector P and the other projectors P_j are captured at the camera position c_i is given by

$$H_{P_i2P} \propto H_{C2P} \times H_{P_i2C} . \quad (1)$$

2.3 Distortion correction

The simple coordinate unification in homography that creates a projected image results in distortions. To address this problem, we apply distortion correction using a pixel map to the projection images. First, we create a new desired camera image C_i' for each camera view, and then obtain a distortion-compensated projection image P_j' using a pixel map C_i2P_j . This correction compensates for the distortions triggered by optical aberrations and the inflatness of the land shape.

3 RESULTS AND DISCUSSION OF GEOMETRIC CALIBRATION

We conducted experiments using nine system projectors (with a resolutions of 1280×800 pixels) and a color camera (with a resolution of 1280×800 pixels) adopted for calibration. For evaluation, we compared the proposed method with homography-based methods [1]. Reference homography H_{W2C_1} and pixel maps C_i2P_j were obtained using a temporally-positioned camera using Gray-code projections.

Figure 4 (a) presents the simulation results using unified projector coordinates via homography, while Figure 4(b) presents the results obtained by the proposed method. A fatal distortion in the radiotelescope using simple homography can be compensated by the proposed method.

Figure 5 presents the averages of the absolute differences in the x- and y-directions between the desired and obtained images at Step 4. The results indicate that our method achieved less error than simple homography, with an average error reduction 84.7%.

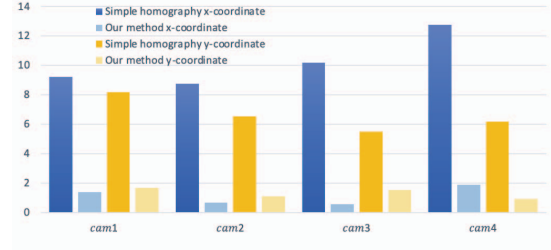
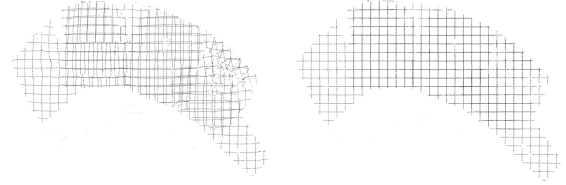


Figure 5: Absolute difference from desired camera image



(a) Simple homography (b) Our method

Figure 6: Alignment simulation with nine projections

Figure 6 presents a combination of the results for the nine projections. Our method exhibits successively corrected results over the entire projection area.

4 CONCLUSION

In this study, we proposed a method to enable large-scale geometrical projection calibration for multi-projector systems on a near-planar arbitrarily-shaped ground. First, the world coordinate system was defined, and the desired camera images were created using homography. To reduce the distortion in the projected image, we employed distortion correction using a pixelmap, in addition to the conventional homography-based alignment.

In the experiment, we simulated how the homography-based method and proposed methods were captured. The results obtained by unifying the coordinate system with homography exhibited large distortions in the radiotelescope area, which were not flat. However, the proposed method reduced the distortion by 84.7% on average. By using camera images placed at each viewing position, the projection was seamlessly connected. In the future, rather than conducting simulations, it will be necessary to investigate and evaluate how it is projected onto the ground.

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